General data organisation

Christian Torp-Pedersen Mikkel Porsborg Andersen Cæcilie Stilling Denholt

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You should use **bef** and **fain** to define the Danish Population, as we do not deliver a predefined population.

Data to projects are provided from many sources, the most important being Statistics Denmark (DST) and the Danish Health Data Authorities (Sundhedsdatastyrelsen, SDS). Data from these sources are by DST conceived as "base data" (Grunddata) and are systematically placed in a folder named **Grunddata**.

All other data will appear in a folder named **Eksterne data**. This folder may be further organised depending on the project. Please be aware that datasets stored in this folder may require additional approvals before they can be used to ensure compliance with current legislation. Therefore, if you have used data from this folder, you should contact the project administrators before submitting any work for publication.

The current listing of dataset is very general and the project you are working on may contain fewer datasets or additional datasets.

This document provides an overview of the structure of **Grunddata**. An important principle for the structure of data has been to change as little as possible from the original data supplied. The reason for this is that it allows all users to interrogate DST and SDS for description of registers and definitions of variables. These institutions have on their web-pages descriptions of datasets and well as variable descriptions.

Register details from the Danish Health Data Authority (Sundhedsdatastyrelsen) is found here: https://sundhedsdatastyrelsen.dk/data-og-registre/nationale-sundhedsregistre Register details from the Statistics Denmark is found here: https://www.dst.dk/extranet/forskningvariabellister/Oversigt%20over%20registre.html

A full list of all external data including variable explanations is found on DDV (Danmarks DataVindue).

Data management with the new structure will result in changes of many programs. To ease this process we have written a SAS and R program to read data in a structured way. These programs are found in V:/Alle/skeleton and are also visible on https://github.com/ctpteam/DST/tree/master/Programming%20guidance/Skeleton.

One very important change for many users is that we provide multiple files for each register according to the DST habit of having one file per year. This change is made to ease data organisation as many users only need to interrogate data from selected years. It further can help to increase speed of data processing by parallising calculations - the document **Parallel processing with SAS and R** describes this. Also, the programs for SAS/R enabling data management are available in the folder V:/data/alle/skeleton

From the late-summer 2026, we will be delivering **parquet files** to most project folders and exclusively to some. Parquet is a column-oriented data format (unlike SAS which is row-oriented). This reduces storage, makes data access faster, and lowers memory consumption. The parquet datasets are divided into multiple smaller parquet files, enabling faster and more efficient data access. We recommend using DuckDB in R to work with these files. DuckDB allows researchers to select variables, filter observations, and combine datasets without loading all data into memory. This improves performance and reduces memory consumption on our shared server. Example programs demonstrating how to work with parquet files using DuckDB are available as a **Skeleton** file.

The following headlines correspond to the subfolders in **Grunddata** on projects.

Setup files

Data in Grunddata are constructed with one or more SAS programs from raw data. Theses SAS programs are available in various versions in this folder.

Cancer

This includes the cancer registry from SDS. Previously, it was called **t_tumor**, but now it contains several files. **tumor_aarlig** refers closest to the old file, and **dimcancergruppering_icd10** references this file.

We now also receive **tumor_aarlig_norcan** which contains the NORDCAN database, and **dimcancergruppering_nordcan** references this.

Death

t_dodsaarsag_1 is the national cause of death register until 2001 when new death certificates were introduced

t_dodsaarsag_2 is the second cause of death register that follows. Note that attempts have been made to provide simple translations of new to olds cause of death definition, but this should be discouraged. It is important to note the update of causes of death is very much delayed.

doedsaarsag_3 is the newest addition to the COD register, beginning in 2022.

Simple registration of death is in the **dod** table. This table combines information from DST (dod) and SDS (t_person) and is updated as much as other data in the project. This dataset only contains date of death for deceased and is not described elsewhere.

Krim

This folder contains information on criminal records and and related subjects. Only available in selected folders.

Laboratory

Previously, this folder held **lab_forsker** and **lab_labidcodes**. After a recent update from SDS, we now receive this data as **laboratorieproevesvar** and **dimlaboratoriekoder**.

NPU codes corresponding to tests are found in V:/data/alle/blodprøver.

Historically we have collected laboratory results from willing sources and these are placed in separate subdirectories corresponding to the old municipal of Copenhagen (Kbh amt), Copenhagen general practitioners laboratory (KPLL), North Region of Denmark and Roskilde. The advantage of these datasets is that they reach further back in time than lab_forsker.

When relevant the pathology register of microscope examinations is also supplied in this directory. Note, that this was previously delivered as **pato** but is now delivered as **fctrekvisition**, **fctpatologiskprocedure** & **dimpatologiskdiagnose**.

LPR

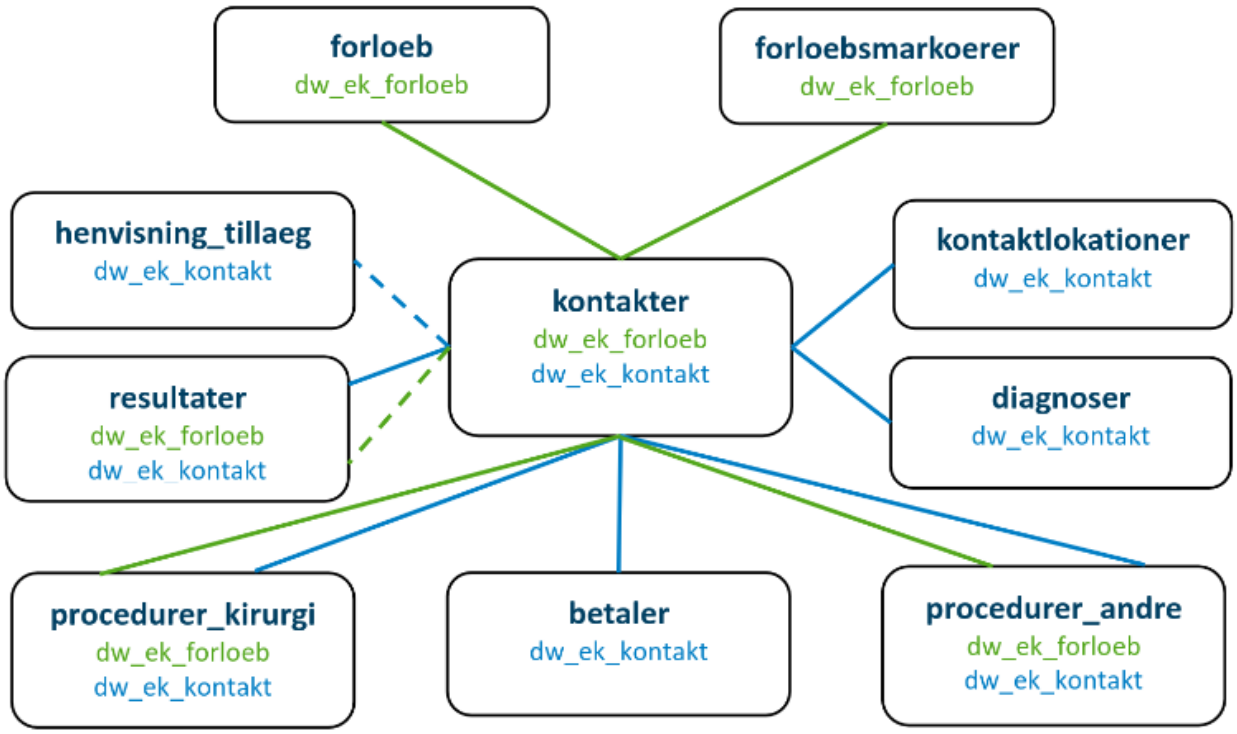
LPR1 uses ICD8-codes and was 1994 replaced with LPR2 and ICD-10 codes. This continued until about March 2019 where LPR3 started. LPR3 is structured very differently from LPR1/2. It has therefore been decided to maintain the partially digested LPR1/2 data in the following deliveries: **diag_indl** contains all diagnoses and selected administrative data. **opr** includes all procedure codes and all examination codes (sksube) - again with added administrative data.

Some projects will also include **opr_old** which are ICD-8 procedures prior to approximately 1994. Other projects also have the file **lpr_bes** which are outpatient contacts from lpr2.

Psychiatric admissions were provided in independent tables. Note than occasionally psychiatric diagnoses appear in the somatic data, but not consistently.

LPR3 datasets are supplied without changes for the populations relevant for the project. Note that we have extensively reduced the number of datasets and variables provided not to overwhelm our servers. Basically the following four tables are made available: kontakter, diagnoser, procedurer_kirurgi and procedurer_andre. For each table the number of variables has been substantially reduced and those interested in those removed need to request them for specific projects.

Those accustomes to LPR1/2 will be searching for diagnosis codes and procedure codes. Of note in LPR3 is is not uncommon to find these fields with "TUL.." codes or other non-standard codes. Some of these refer to side of patients (left/right) and for others we do not know the meaning. But when strange codes are used you will find that the field "..._parent" has a more ordinary codes.



LPR3 contains the datasets provided in the figure. Additionally the key variables to connect data are shown.

For a short introduction to LPR3 please see: https://github.com/ctpteam/DST/blob/master/Guide%20to%20LPR3/Vejledning%20til%20LPR3_F.pdf For a thorough introduction to LPR3 see: https://github.com/ctpteam/DST/blob/master/Guide%20to%20LPR3/LPR_Indberetningsvejledning_lang.pdf

Medication

This directory includes all **lmdb** files available. Additional data regarding compounds can be found in **laegelmiddeloplysninger**, which in various updated versions are available on **V:/data/Alle/LMDBdata**

Be careful when loading large sequences of data. We routinely obtain datasets with all prescriptions during a year, but sometimes also prescriptions covering the first 6 months. If both are available and included it will result in multiple duplicate prescriptions that can disturb calciulations of therapy periods.

Mfr

mfr is the national birth register and this dataset has data from 1997-2019.

We have older data also **mfr__lfoed** - born alive prior to 1997 **mfr__dfoed** - stillborn prior to 1997 **nydfoed__2010** and **nylfoed__2010** are also delivered and may not provide new data.

With the introduction of LPR3 all births from 2019 are found in the LPR3-table **nyfoedte**.

Nursing home

The dataset **plhjem** is the data we have from nursing homes generated by DST by request from us and using the number of old people at an address to qualify an address to be examined as to whether is was a nursing home.

From 2016 the more official data **aepi** has nursing home data.

The registers **aefv**, **aelh**, **aeph** and **aetr** are various types of personal assistance

Population

The main population register is the **bef** tables for each year since 1985 and from 2008 each quarter of a year. This is the main official register to define danish residents at any time.

Prior to 1985 the **fain** tables provide some of the data found in **bef**.

In addition the dataset **sexBirth** provides date of birth and sex (0=female,1=male) available from the cpr-register (t_person). This particular dataset is not described further elsewhere.

Projects may or may not directly have the **t__person** register. Not that this register also includes legal parents and "C_STATUS" which is 90 for death and also includes information on migration and people disappeared.

Country of origin is found in the **iepe** dataset and all migrations into and away from Denmark are found in the **vnds** table.

Social

The tables **indXXXX** (where Xs represent years) include income and wealth of individuals. This table also includes individual income adjusted for family composition. The table does not adjust for inflation.

The tables **uddaXXXX** have maximum level of education for all individuals. There are SAS formats to translate to understandable values and also an R-function in heaven to do this.

The table **uddfXXXX** also contains maximum level of education. Only the last year is necesssary to include as it is updated with new data annually. It appears to include maximum education for more people than udda-files, perhaps because imported educations are also registered here.

The table **dreamXXXXXX** is an updated version of dream that for each week has codes indicating public support and for each month indication of association to particular working areas.

Some projects include **kotre** which is a student register. Further educational registers may be available in selected projects.

SSS

SYSI and **SSSY** are data from practitioners to many types representing different time periods.

Other data

For all projects there is a folder named **Ekstern Data** (external data). The data in this folder can be found in the description files for the project - named variables and datasets. Further explanations can be found in the approvals from DST and SDS or on DDV.

V:/data/Alle

On the V-path subfolder "Alle" is a number of moderately organised folders for a variety of anonymous and useful data. Not in particular details of medication in LMDBdata, text versions of diagnoses in ICD-8 and ICD-10

Note also here the "Skeleton" folder that includes sample programs to import data from the structure required by Statistics Denmark

DST formats

On the desktop of Statistics Denmark there is a **STAR** called formats. This provides access to multiple format that can be helpful for classifying complex codes. For example, the four-digit codes representing the highest level of completed education can be used to classify educational attainment according to the International Standard Classification of Education (ISCED).